

Intelligent Campus Highlight Board System

Project Running Guide (Windows)

This document explains the complete steps required to run the Intelligent Campus Highlight Board System project on Windows.

1. Prerequisites

Flutter SDK installed
Android Studio / Emulator or connected Android device
Python 3.11+ installed
Git installed
MongoDB running locally or MongoDB Atlas available

2. Backend Setup

Go to backend folder:
`cd F:\softwareEng\backend`

Create virtual environment:
`python -m venv .venv`

Activate virtual environment:
`.venv\Scripts\activate`

Install Python dependencies:
`pip install -r requirements.txt`

Create environment file:
`copy .env.example .env`

Edit .env and configure:
MONGODB_URI
MONGODB_DB_NAME
optionally FIREBASE_CREDENTIALS_PATH
optionally CORS_ALLOW_ALL=true

If MongoDB is not installed locally, run:
`docker compose up -d`

3. Seed Demo Data

Run the following command:
`python scripts\seed_data.py`

4. Start Backend Server

Run backend server:
`uvicorn app.main:app --host 0.0.0.0 --port 8010 --reload`

Verify backend URLs:
`http://127.0.0.1:8010/health`
`http://127.0.0.1:8010/docs`

5. Flutter App Setup

```
Go to project root:
cd F:\SoftwareEng

Install Flutter dependencies:
flutter pub get

Run Flutter app:
flutter run

For Android Emulator use API URL:
http://10.0.2.2:8010/api/v1

For Real Device:
- Start backend with --host 0.0.0.0
- Find PC LAN IP
- Use:
http://<YOUR_PC_IP>:8010/api/v1
```

6. Login Credentials

```
Student:
student@campus.edu / Student@123

Faculty:
faculty@campus.edu / Faculty@123

Admin:
admin@campus.edu / Admin@123
```

7. Important Notes

```
The Flutter app initializes Firebase if configured.
.env file is ignored by git for security.
If using Flutter Web, set CORS_ALLOW_ALL=true in .env.
```

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